

Disproof of TLMC Conjecture 00000001068

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Verdict

FALSE. The conjecture fails already at $p = 11$.

1 The conjecture

The maximal sum-free subsets of $\mathbb{F}_p$ are exactly the intervals $((p+1)/3, 2(p-1)/3)$, uniquely up to dilation (the complete Diananda–Yap characterization over finite fields).

A set $A \subseteq \mathbb{F}_p$ is *sum-free* if $(A+A) \cap A = \emptyset$, and *maximal* sum-free if it is not properly contained in another sum-free set.

2 Counterexample: $p = 11$

Theorem 1. *In $\mathbb{F}_{11}$ the set $A = \{4, 5, 6, 7\}$ is a maximal sum-free subset with $|A| = 4$, whereas the conjectured interval $I = ((p+1)/3, 2(p-1)/3) = (4, 20/3) = \{5, 6\}$ has $|I| = 2$ and is not maximal. Hence the conjecture is false.*

Proof. A is sum-free. A direct computation of all sums of two elements of A gives

$$A + A = \{0, 1, 2, 3, 8, 9, 10\} = \mathbb{F}_{11} \setminus A,$$

which is disjoint from A .

A is maximal. Every element outside A is a sum of two elements of A :

$$0 = 5 + 6, \quad 1 = 5 + 7, \quad 2 = 6 + 7, \quad 3 = 7 + 7, \quad 8 = 4 + 4, \quad 9 = 4 + 5, \quad 10 = 4 + 6.$$

If $x \notin A$ were added to A , then $x \in A + A \subseteq (A \cup \{x\}) + (A \cup \{x\})$ would meet $A \cup \{x\}$, so no enlargement of A is sum-free.

I is not maximal. By definition of the open interval with exact rational endpoints, $x \in I$ iff $3x > 12$ and $3x < 20$, i.e. $I = \{5, 6\}$. Since $I \subsetneq A$ and A is sum-free, I is not maximal. The same holds for every nonzero dilate: for $d = 1, \dots, 10$ the dilate $dI = \{5d, 6d\} \bmod 11$ has exactly two elements and satisfies $dI \subsetneq dA$, where dA is sum-free (dilation by a unit preserves sum-freeness, and this is directly verified for all d); the element $4d \bmod 11$ lies in $dA \setminus dI$. Thus the family conjectured to be “the maximal sum-free subsets” contains no maximal sum-free set at $p = 11$.

A lies outside the conjectured family. A is maximal sum-free by the above, and every dilate of I has 2 elements while $|A| = 4$; so A is not a dilate of I . $\square$

Both inclusions of the claimed characterization therefore fail: the conjectured family contains no maximal sum-free set, and there exists a maximal sum-free set outside it.

3 Computational verification

A brute-force census over all 2^{11} subsets of $\mathbb{F}_{11}$ confirms the maximum sum-free size is $4 = \lfloor (p+1)/3 \rfloor$ and that there are exactly 15 maximal sum-free subsets, with sizes in $\{3, 4\}$; $A = \{4, 5, 6, 7\}$ is one of the size-4 maximal sets. All numbers above are recomputed by the accompanying script `reproduce.py` and are machine-checked in pure Lean 4 (no axioms, no `sorry`; see `lean4/`).

4 Boundary remarks

- The attack is robust to endpoint conventions: with integer-division endpoints the open reading gives $(4, 6) = \{5\}$ and the closed reading gives $[4, 6] = \{4, 5, 6\}$; both are still strictly contained in the sum-free set A , hence non-maximal under every reading.
- The breakage is not specific to $p = 11$. Under the exact-rational open reading, $|I|$ equals $0, 0, 0, 1, 2, 3$ for $p = 2, 3, 5, 7, 11, 13$, while the true maximum sum-free size equals $1, 1, 2, 2, 4, 4$; the claimed interval is non-maximal for every prime in this census.
- The genuine Diananda–Yap extremal interval is $\{\lfloor p/3 \rfloor + 1, \dots, \lfloor 2p/3 \rfloor\}$, equal to $\{4, \dots, 7\}$ for $p = 11$. The conjecture’s open endpoints $((p+1)/3, 2(p-1)/3)$ truncate that interval, which is precisely what destroys maximality. A single counterexample refutes the universal claim.